

SCHMUECKE, Germany

WMO: 105520

Lat: 50.6544N

Lon: 10.7694E

Elev: 939

StdP: 90.54

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-16.1	-12.9	-18.6	0.8	-12.1	-15.9	1.1	-11.8	17.2	3.5	14.8	0.8	6.5	50	0.766

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.4	23.6	16.0	21.9	15.4	20.2	14.6	17.1	21.6	16.1	20.5	15.2	19.1	4.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.4	12.3	18.5	14.4	11.5	17.5	13.6	10.9	16.8	51.8	21.6	48.6	20.7	45.9	19.0	20.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.9	12.2	10.8	DB	-15.9	27.1	5.0	1.4	-19.5	28.2	-22.4	29.0	-25.2	29.8	-28.9	30.9	(4)
(5)			WB	-16.1	19.4	4.9	0.9	-19.6	20.0	-22.5	20.6	-25.3	21.1	-28.9	21.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.9	-3.4	-4.1	-0.2	3.2	8.8	11.1	13.8	13.6	10.1	6.1	0.7	-1.7	(6)
(7)		DBStd	7.51	4.68	4.93	4.27	4.26	4.14	4.15	3.90	3.65	3.67	3.86	4.09	3.64	(7)
(8)		HDD10.0	2277	416	394	316	208	74	35	9	9	44	132	278	362	(8)
(9)		HDD18.3	4921	674	628	573	455	297	219	148	152	248	379	528	620	(9)
(10)		CDD10.0	412	0	0	1	3	36	69	126	120	46	12	0	0	(10)
(11)		CDD18.3	14	0	0	0	0	0	3	7	5	0	0	0	0	(11)
(12)		CDH23.3	57	0	0	0	0	1	8	29	18	0	0	0	0	(12)
(13)		CDH26.7	3	0	0	0	0	0	0	2	1	0	0	0	0	(13)
(14)	Wind	WSAvg	5.3	6.6	5.9	5.9	4.8	4.7	4.6	4.4	4.5	5.1	5.7	5.7	5.8	(14)
(15)	Precipitation	PrecAvg	914	83	70	69	55	73	75	100	70	78	68	82	93	(15)
(16)		PrecMax	1181	181	184	140	131	225	131	174	189	170	217	148	237	(16)
(17)		PrecMin	652	2	11	15	6	15	18	37	19	15	21	1	26	(17)
(18)		PrecStd	135	40	39	38	27	45	32	31	33	35	41	36	43	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.6	8.5	13.7	16.1	21.0	24.6	26.4	26.3	22.0	18.2	12.2	8.1	(19)
(20)			MCWB	4.3	5.7	7.7	9.2	13.3	17.0	17.4	17.2	14.7	12.5	6.8	6.0	(20)
(21)		2%	DB	4.4	6.2	10.0	13.7	19.0	21.9	23.7	23.1	19.5	15.3	9.2	5.8	(21)
(22)			MCWB	2.9	4.6	6.5	8.5	12.5	15.4	16.0	16.0	14.7	11.3	6.1	4.7	(22)
(23)		5%	DB	3.0	4.2	7.5	11.7	17.1	19.7	22.0	21.3	17.4	13.0	7.9	4.6	(23)
(24)			MCWB	2.4	2.9	5.1	7.4	11.6	14.2	15.7	15.2	13.8	10.3	5.7	3.7	(24)
(25)		10%	DB	1.8	2.0	5.1	9.5	15.2	17.6	20.3	19.3	15.4	11.2	6.5	3.2	(25)
(26)			MCWB	1.3	1.0	4.0	6.2	10.6	12.9	15.0	14.4	12.6	9.5	4.9	2.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.3	6.5	8.3	10.4	14.4	17.7	18.6	17.8	16.8	13.7	8.7	6.9	(27)
(28)			MCDWB	5.7	7.7	11.2	14.1	19.0	21.9	24.1	24.2	19.0	16.7	9.7	7.1	(28)
(29)		2%	WB	3.5	4.9	6.8	8.9	13.1	16.0	17.4	16.6	15.3	12.1	7.4	5.3	(29)
(30)			MCDWB	3.7	5.8	9.7	13.3	17.0	20.5	21.9	21.5	18.3	13.9	8.3	5.5	(30)
(31)		5%	WB	2.4	2.9	5.4	7.6	12.0	14.9	16.2	15.7	14.1	10.9	6.4	3.9	(31)
(32)			MCDWB	2.7	3.8	7.1	11.0	16.1	18.7	20.6	20.2	16.6	12.8	7.4	4.2	(32)
(33)		10%	WB	1.4	1.1	4.1	6.6	11.1	13.7	15.3	14.8	13.2	9.9	5.0	2.7	(33)
(34)			MCDWB	1.8	1.9	5.1	9.3	14.9	17.0	19.2	18.5	15.1	11.0	6.3	3.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.1	4.6	4.6	6.2	7.1	6.6	7.4	6.9	5.8	5.0	3.9	3.5	(35)
(36)			MCDBR	5.0	6.5	7.6	9.3	9.7	9.3	9.7	9.6	8.4	7.6	6.7	5.1	(36)
(37)			MCWBR	4.6	4.7	4.8	5.5	5.5	5.2	5.0	5.0	5.0	4.8	4.9	4.5	(37)
(38)		5% WB	MCDWR	4.8	5.8	6.8	8.7	8.7	8.3	8.9	8.3	7.4	6.2	5.8	4.4	(38)
(39)			MCWBR	4.6	4.6	4.8	5.6	5.3	5.0	5.1	4.8	4.9	4.3	4.7	4.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.278	0.298	0.332	0.365	0.369	0.367	0.379	0.373	0.344	0.322	0.301	0.275	(40)	
(41)		taud	2.451	2.418	2.379	2.322	2.330	2.345	2.309	2.350	2.422	2.484	2.484	2.448	(41)	
(42)		Ebn at Noon	773	845	872	876	886	888	870	858	848	805	734	718	(42)	
(43)		Edh at Noon	58	78	97	115	120	119	121	111	92	73	57	51	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.77	1.50	2.56	4.06	4.85	5.28	5.01	4.32	3.10	1.81	0.85	0.58	(44)	
(45)		RadStd	0.14	0.23	0.34	0.52	0.57	0.45	0.60	0.44	0.43	0.22	0.12	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47) Regional (41 neighbors)		N/A	N/A	N/A	N/A	N/A	+0.30	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page